

Analytic Geometry

chapter 11: Straight Lines

• Review of basic concepts:

① Distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

② Mid-point formula:

$$M(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

③ Section- Formula:

a) Internal Division:

$$(x, y) = \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

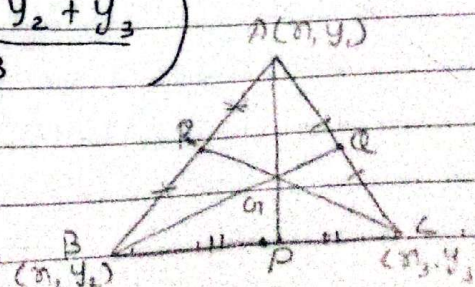
b) External Division

$$(x, y) = \left(\frac{m_1 x_2 - m_2 x_1}{m_1 - m_2}, \frac{m_1 y_2 - m_2 y_1}{m_1 - m_2} \right)$$

④ Centroid of triangle

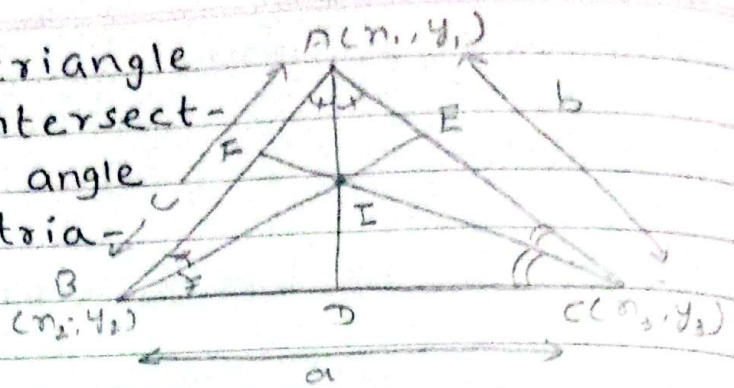
→ The point of intersection of Median of a triangle is called its centroid.

$$(x, y) = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$



⑤ Incentre of triangle

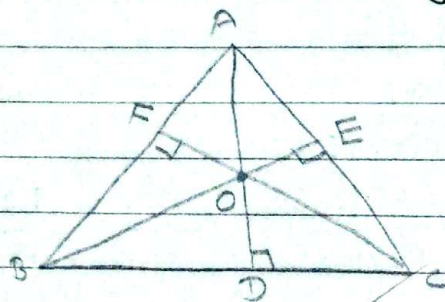
↳ The point of intersection of internal angle bisectors of a triangle is called its incentre.



Co-ordinates of incentre (x, y)

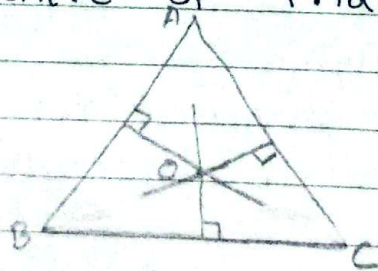
$$(x, y) = \left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c} \right)$$

⑥ Orthocentre of triangle



↳ The point of intersection of altitudes of a triangle is called its orthocentre.

⑦ Circumcentre of triangle



↳ The point of intersection of perpendicular bisectors of sides of triangle is called its Circumcentre.

⑧ Area of Triangle

→ Area of triangle having vertices (x_1, y_1) , (x_2, y_2) and (x_3, y_3) is

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix}$$

$$= \frac{1}{2} [(x_1 y_2 - x_2 y_1) + (x_2 y_3 - x_3 y_2) + (x_3 y_1 - y_3 x_1)]$$

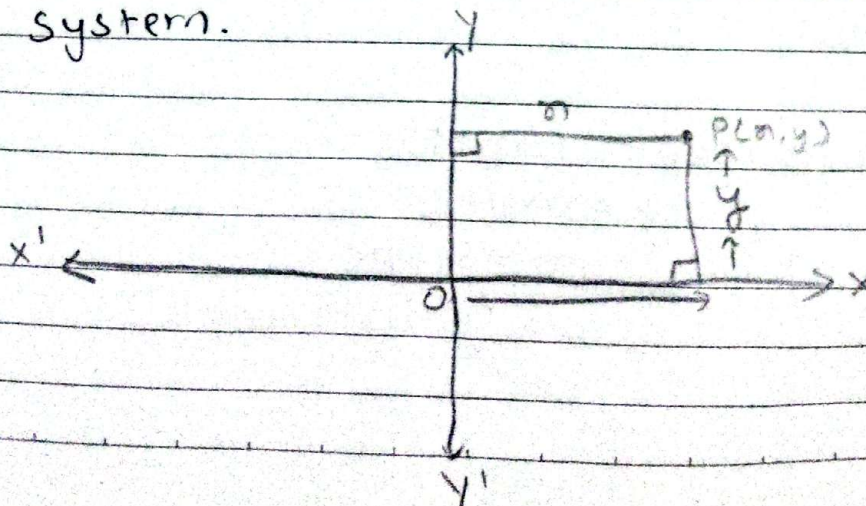
OR

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

⑨ Locus

→ Locus is a path followed by a moving point under certain condition. It is represented by an equation.

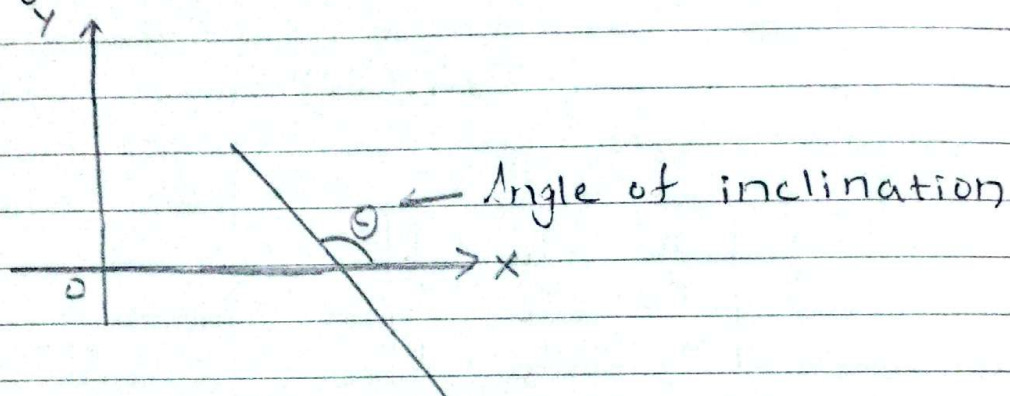
• 2 Dimensional / Cartesian coordinate system.



Straight line and slope

- Straight line - Locus of moving point on a plane and is described by the equation of the form $ax + by + c = 0$

- Angle of inclination



- Slope: tangent of angle of inclination $= \tan \theta$

$$\text{Slope}(m) = \tan \theta = \frac{y_2 - y_1}{x_2 - x_1} = \frac{x \text{ coe.}}{y \text{ coe.}}$$

Different forms of equation of straight line:

- ① Slope-intercept form:

$$y = mx + c$$

Where, $m = \text{Slope}$

$c = y\text{-intercept}$

② Double-intercept form:

$$\frac{x}{a} + \frac{y}{b} = 1$$

where,

$a = x$ -intercept

$b = y$ -intercept.

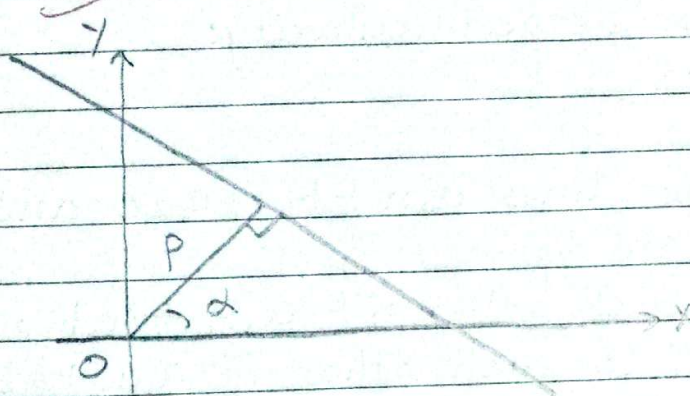
③ Normal / perpendicular form:

$$x \cos \alpha + y \sin \alpha = P$$

Where,

$P =$ length of perpendicular from origin

$\alpha =$ Angle of inclination of perpendicular



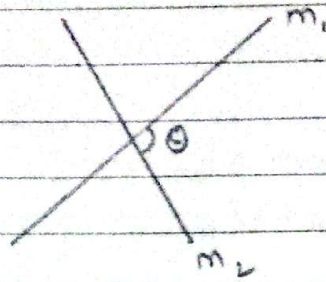
④ Point - Slope form:

$$y - y_1 = m(x - x_1)$$

⑤ General form:

$$ax + by + c = 0$$

Angle between two straight lines:



$$> \tan \theta = \pm \frac{m_1 - m_2}{1 + m_1 \cdot m_2}$$

• Condition for parallelism:
 $m_1 = m_2$

• Condition for perpendicularity:
 $m_1 \cdot m_2 = -1$

Q. Show that the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are

- i) Parallel to each other if $a_1b_2 - a_2b_1 = 0$
- ii) Perpendicular to each other if $a_1a_2 + b_1b_2 = 0$

i) Soln,

$$m_1 = \frac{-a_1}{b_1}$$

$$m_2 = \frac{-a_2}{b_2}$$

~~For~~ $\theta = 180^\circ$

Now,

$$\tan \theta = \pm \frac{m_1 - m_2}{1 + m_1 \cdot m_2}$$

$$\text{or, } \tan 180 = \pm \frac{m_1 - m_2}{1 + m_1 \cdot m_2}$$

or, $0 = \pm m_1 - m_2$

or, $0 = \pm \left[\frac{-a_1}{b_1} - \left(\frac{-a_2}{b_2} \right) \right]$

or, Taking $\oplus$ ve

$$\frac{-a_1}{b_1} + \frac{a_2}{b_2} = 0 \Rightarrow \frac{a_1}{b_1} - \frac{a_2}{b_2} = 0 \Rightarrow \frac{a_1}{b_1} = \frac{a_2}{b_2}$$

$$\therefore a_1 b_2 - a_2 b_1 = 0 \quad \checkmark$$

Taking $\ominus$ ve

$$\frac{a_1}{b_1} - \frac{a_2}{b_2} = 0 \Rightarrow \frac{a_1}{b_1} = \frac{a_2}{b_2} \Rightarrow a_1 b_2 - a_2 b_1 = 0 \quad \checkmark$$

ii) $\theta = 90^\circ$

Now,

$$\tan 90 = \frac{m_1 - m_2}{1 + m_1 \cdot m_2}$$

$$\frac{1}{0} = \frac{m_1 - m_2}{1 + m_1 \cdot m_2}$$

$$\text{or, } 1 + m_1 \cdot m_2 = 0$$

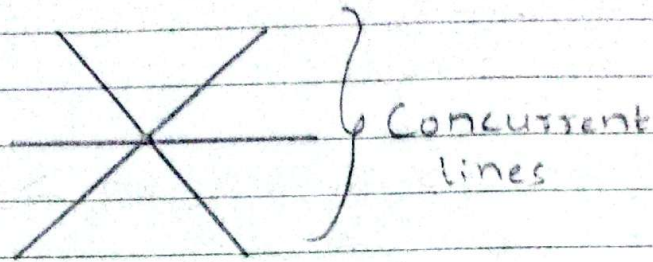
$$\text{or, } 1 + \left(\frac{-a_1}{b_1} \right) \left(\frac{-a_2}{b_2} \right) = 0$$

$$\text{or, } 1 + \frac{a_1 a_2}{b_1 b_2} = 0$$

$$\text{or, } b_1 b_2 + a_1 a_2 = 0$$

$$\text{or, } a_1 a_2 + b_1 b_2 = 0 \quad \checkmark$$

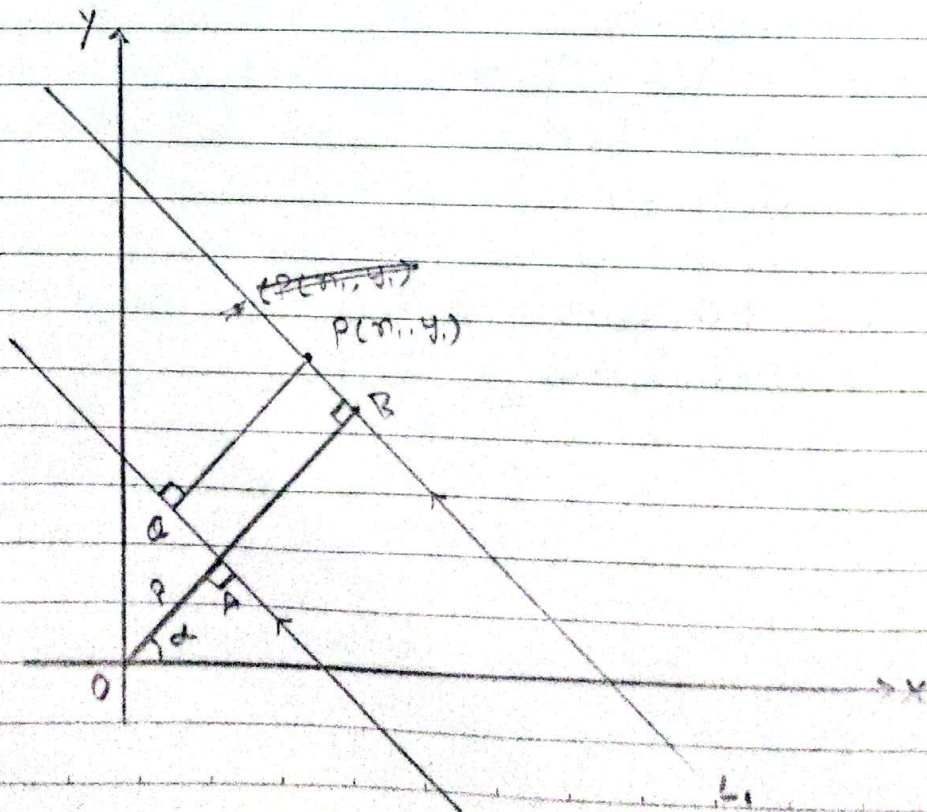
Concurrent lines



→ If three or more lines are intersecting at a point then these lines are called concurrent lines and the point of intersection is called point of concurrency.

Length of perpendicular

Consider a line L having equation $x \cos \alpha + y \sin \alpha = p$ and a point $P(x_1, y_1)$.



Draw $PQ \perp L$ and $OA \perp L$, then

$$OA = P \text{ and } \angle OAx = \alpha$$

Again, draw a line L_1 passing through the point P and parallel to L . and then produce OA to meet the line L_1 at point B . Then $OB \perp L_1$.

Now, the equation of line L_1 is

$$x \cos \alpha + y \sin \alpha = OB$$

This line passes through the point $P(x_1, y_1)$. So,

$$x_1 \cos \alpha + y_1 \sin \alpha = OB$$

Finally,

$$PQ = AB$$

$$= |OB - OA|$$

$$= |x_1 \cos \alpha + y_1 \sin \alpha - P|$$

$\therefore$ The length of perpendicular drawn from a point (x_1, y_1) to the line $x \cos \alpha + y \sin \alpha = P$ is $|x_1 \cos \alpha + y_1 \sin \alpha - P|$.

Note:

↳ For the equation $ax + by + c = 0$, the corresponding normal form is $ax + by = -c$

$ax + by = -c$
 Dividing by $\pm \sqrt{a^2 + b^2}$ on both sides,

$$\frac{a}{\pm \sqrt{a^2+b^2}}x + \frac{b}{\pm \sqrt{a^2+b^2}}y = \frac{-c}{\pm \sqrt{a^2+b^2}}$$

Comparing with $x \cos \alpha + y \sin \alpha = p$

$$\cos \alpha = \frac{a}{\pm \sqrt{a^2+b^2}}$$

$$\sin \alpha = \frac{b}{\pm \sqrt{a^2+b^2}}$$

$$p = \frac{-c}{\pm \sqrt{a^2+b^2}}$$

Now,

$$\begin{aligned} \text{length of perpendicular} &= |x \cos \alpha + y \sin \alpha - p| \\ &= \left| x \times \frac{a}{\pm \sqrt{a^2+b^2}} + y \times \frac{b}{\pm \sqrt{a^2+b^2}} + \frac{c}{\pm \sqrt{a^2+b^2}} \right| \\ &= \left| \frac{ax + by + c}{\sqrt{a^2+b^2}} \right| \end{aligned}$$

→ Distance between two parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is $\left| \frac{c_1 - c_2}{\sqrt{a^2+b^2}} \right|$

→ Given a line $ax + by + c = 0$, then

i) Parallel line: $ax + by + k = 0$

ii) Perpendicular line: $bx - ay + k = 0$

} Where k is constant

Exercise - 11

① Find the length perpendicular from the point P to line L.

$$\hookrightarrow P(a, b); L: \frac{x}{a} + \frac{y}{b} = 1$$

$\rightarrow$ Given,

$$P(a, b) = (x_1, y_1)$$

$$\text{and a line } L: \frac{x}{a} + \frac{y}{b} = 1$$

$$\text{i.e. } \frac{bx + ay}{ab} = 1$$

$$\text{i.e. } bx + ay = ab$$

$$\text{i.e. } bx + ay - ab = 0$$

We have,

$$\text{length of perpendicular} = \left| \frac{bx_1 + ay_1 - ab}{\sqrt{b^2 + a^2}} \right|$$

$$= \frac{\pm ab + ab - ab}{\sqrt{a^2 + b^2}}$$

$$= \frac{\pm ab}{\sqrt{a^2 + b^2}} \text{ units} \quad \perp$$

Q. Find the distance between two parallel lines $4x - 3y + 8 = 0$ and $8x - 6y - 14 = 0$

$\rightarrow$ Soln,

Given parallel lines are,

$$4x - 3y + 8 = 0 \rightarrow (i)$$

$$8x - 6y - 14 = 0$$

Dividing on both sides,
 $4x - 3y - 7 = 0 \rightarrow (ii)$

Here,

$$a=4, b=-3, c_1=8 \text{ and } c_2=-7$$

Distance between parallel lines

$$= \pm \frac{c_1 - c_2}{\sqrt{a^2 + b^2}}$$

$$= \pm \frac{8 - (-7)}{\sqrt{4^2 + (-3)^2}}$$

$$= \pm \frac{8 + 7}{\sqrt{16 + 9}}$$

$$= \pm \frac{15}{\sqrt{25}}$$

$$= \pm \frac{15}{5}$$

$$= \pm 3$$

$$= 3 \text{ units}$$

④ If P is the length of perpendicular from the point $(1, c)$ to the line $y = mx + c$ then show that $\frac{1}{p^2} - \frac{1}{m^2} = 1$

→ Soln,

Given equation is,

$$y = mx + c$$

$$mx - y + c = 0$$

$$\text{Point} = (1, c) = x_1, y_1$$

length of perpendicular = P

Now,

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$$\text{length of perpendicular} = \pm \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}$$

$$\text{or, } P = \pm \frac{mx_1 - y_1 + c}{\sqrt{m^2 + 1^2}}$$

$$\text{or, } P = \pm \frac{m \cdot 1 - 1 + c}{\sqrt{m^2 + 1}}$$

$$\text{or, } P = \pm \frac{m}{\sqrt{m^2 + 1}}$$

Squaring on both side,

$$P^2 = \frac{m^2}{m^2 + 1}$$

$$\text{or, } \frac{1}{P^2} = \frac{m^2 + 1}{m^2}$$

$$\text{or, } \frac{1}{P^2} = \frac{m^2}{m^2} + \frac{1}{m^2}$$

$$\text{or, } \frac{1}{P^2} = 1 + \frac{1}{m^2}$$

$$\text{or, } \frac{1}{P^2} - \frac{1}{m^2} = 1$$

$$\boxed{\therefore \frac{1}{P^2} - \frac{1}{m^2} = 1} \quad \text{X}$$

1.

a) $P(1,2)$; $L: 3x - 2y + 4 = 0$

→ Given,

$$P(1,2) = (x_1, y_1)$$

and a line $L: 3x - 2y + 4 = 0$

We have,

$$\begin{aligned} \text{Length of perpendicular} &= \left| \frac{3x_1 - 2y_1 + 4}{\sqrt{3^2 + (-2)^2}} \right| \\ &= \left| \frac{3 \times 1 - 2 \times 2 + 4}{\sqrt{9 + 4}} \right| \\ &= \left| \frac{3 - 4 + 4}{\sqrt{13}} \right| \\ &= \pm \frac{3}{\sqrt{13}} \end{aligned}$$

∴ Length of perpendicular = $\pm \frac{3}{\sqrt{13}}$ ✓

b) $P(0,0)$; $L: x - \sqrt{3}y = 2$

→ Given,

$$P(0,0) = (x, y)$$

and a line $L: x - \sqrt{3}y = 2$

or, $x - \sqrt{3}y - 2 = 0$

We have,

$$\begin{aligned} \text{Length of perpendicular} &= \left| \frac{x - \sqrt{3}y - 2}{\sqrt{1^2 + (-\sqrt{3})^2}} \right| \\ &= \left| \frac{0 - \sqrt{3} \times 0 - 2}{\sqrt{1 + 3}} \right| \\ &= \left| \frac{-2}{\sqrt{4}} \right| \end{aligned}$$

$$= \left| \frac{-2}{2} \right|$$

$$= |-1|$$

$$= 1$$

$\therefore$ length of perpendicular = 1 u.

d) $P(3,4); L: x=3$

$\rightarrow$ Given,

$$P(3,4) = (x, y)$$

and a line $L: x=3$

$$\text{or, } x-3=0$$

We have,

$$\text{length of perpendicular} = \left| \frac{x-3}{\sqrt{1^2+0^2}} \right|$$

$$= \left| \frac{3-3}{\sqrt{1}} \right|$$

$$= \left| \frac{0}{1} \right|$$

$$= 0$$

$\therefore$ length of perpendicular = 0 u.

e) $P(3,4); L: y=6$

$\rightarrow$ Given,

$$P(3,4) = (x, y)$$

and a line $L: y=6 \Rightarrow y-6=0$

We have,

$$\text{length of perpendicular} = \left| \frac{y-6}{\sqrt{0^2+1^2}} \right|$$

$$\begin{aligned}
 &= \left| \frac{4-6}{\sqrt{1}} \right| \\
 &= \left| \frac{-2}{1} \right| \\
 &= |-2| \\
 &= 2
 \end{aligned}$$

$\therefore$ length of perpendicular = 2 $\frac{1}{2}$.

2. Find the distance between the parallel lines.

a) $x+2y=3$ and $x+2y=4$

$\rightarrow$ Soln,

Given lines are,

$$x+2y=3$$

$$x+2y-3=0 \rightarrow (i)$$

$$x+2y=4$$

$$x+2y-4=0 \rightarrow (ii)$$

~~distance~~ $\times$ | Here,
 $a=1, b=2, C_1=-3, C_2=-4$

Distance between parallel lines

$$= \left| \frac{C_1 - C_2}{\sqrt{a^2 + b^2}} \right|$$

$$= \left| \frac{-3 - (-4)}{\sqrt{1^2 + 2^2}} \right|$$

$$= \left| \frac{-3 + 4}{\sqrt{1+4}} \right|$$

$$= \left| \frac{1}{\sqrt{5}} \right|$$

$$= \frac{1}{\sqrt{5}} \text{ units}$$

$\therefore$ Distance between parallel lines $= \frac{1}{\sqrt{5}}$ units.

b) $3x - 4y + 5 = 0$ and $6x - 8y + 7 = 0$

$\rightarrow$ Soln,

Given lines are,

$$3x - 4y + 5 = 0 \rightarrow (i)$$

AND,

$$6x - 8y + 7 = 0$$

$$3x - 4y + \frac{7}{2} = 0 \rightarrow (ii)$$

Here,

$$a = 3, b = -4, C_1 = 5, C_2 = \frac{7}{2}$$

Distance between parallel lines

$$= \left| \frac{C_1 - C_2}{\sqrt{a^2 + b^2}} \right|$$

$$= \left| \frac{5 - \frac{7}{2}}{\sqrt{3^2 + (-4)^2}} \right|$$

$$= \left| \frac{\frac{3}{2}}{\sqrt{9 + 16}} \right|$$

$$= \left| \frac{3}{2} \times \frac{1}{5} \right|$$

$$= \frac{3}{10}$$

$\therefore$ Distance between two parallel lines
 $= \frac{3}{10}$ units.

3. If the length of perpendicular drawn from a point $(k, 3)$ to the line $3x + 4y + 5 = 0$ is 4, then find the value of k .

$\rightarrow$ Soln,

$$P(k, 3) = (x, y)$$

$$\text{length of perpendicular (P)} = 4$$

$$\text{Line : } 3x + 4y + 5 = 0$$

Here,

$$a = 3, b = 4, c = 5$$

We have,

$$\text{length of perpendicular} = \left| \frac{3x + 4y + 5}{\sqrt{3^2 + 4^2}} \right|$$

$$\text{or, } 4 = \pm \frac{3 \times k + 4 \times 3 + 5}{\sqrt{9 + 16}}$$

$$\text{or, } 4 = \pm \frac{3k + 12 + 5}{\sqrt{25}}$$

$$\text{or, } 4 = \pm \frac{3k + 17}{5}$$

$$\text{or, } 20 = \pm 3k + 17$$

Taking $\oplus$ ve sign,
 $20 = 3k + 17$

$$\text{or, } 3k = 20 - 17$$

$$\text{or, } k = \frac{3}{3}$$

$$\therefore k = 1$$

Taking negative sign,

$$20 = -3k - 17$$

$$\text{or, } 3k = -17 - 20$$

$$\text{or, } k = \frac{-37}{3}$$

$$\therefore k = 1, -\frac{37}{3}$$

5. Find the point on x -axis from which the length of perpendicular on the line $y = x$ is a units.

→ Given,

Point on x -axis $(P) = (x, 0) = (x, y)$

line: $y = x$

$$\text{or, } x - y = 0$$

Where, $a = 1, b = -1$

We have,

$$\text{length of perpendicular} = \left| \frac{x - y}{\sqrt{1^2 + (-1)^2}} \right|$$

$$\text{or, } a = \pm \frac{x - 0}{\sqrt{1 + 1}}$$

$$\text{or, } a = \pm \frac{x}{\sqrt{2}}$$

$$\text{or, } x = \pm \sqrt{2} a$$

$\therefore$ Points on x -axis $(P) = (\pm \sqrt{2} a, 0)$

6. What are the points on y -axis, whose perpendicular distance from straight line $\frac{x}{a} + \frac{y}{b} = 1$ is b units?

$\rightarrow$ Soln,

Let points on y -axis are $P(0, y)$
perpendicular distance = b units

$$\text{line: } \frac{x}{a} + \frac{y}{b} = 1$$

$$\text{or, } \frac{bx + ay}{ab} = 1$$

$$\text{or, } bx + ay = ab$$

$$\text{or, } bx + ay - ab = 0$$

Where, $a = b$, $b = a$, $c = -ab$

Now,

$$\text{perpendicular distance} = \left| \frac{bx + ay - ab}{\sqrt{b^2 + a^2}} \right|$$

$$\text{or, } b = \left| \frac{bx + ay - ab}{\sqrt{b^2 + a^2}} \right|$$

$$\text{or, } b = \pm \frac{ay - ab}{\sqrt{b^2 + a^2}}$$

Squaring on both sides

$$\frac{b^2}{1} = \frac{(ay - ab)^2}{a^2 + b^2}$$

$$\text{or, } b \sqrt{b^2 + a^2} = \pm ay - ab$$

$$\text{or, } \pm ay = b\sqrt{b^2 + a^2} + ab$$

$$\text{or, } y = \frac{b(a \pm \sqrt{b^2 + a^2})}{a}$$

$$\text{or, } y = \frac{b}{a} [a \pm \sqrt{b^2 + a^2}]$$

$\therefore$ points on y-axis are $\left[0, \frac{b}{a} [a \pm \sqrt{b^2 + a^2}]\right]$

7. Find the equation of the line parallel to $5x + 12y = 0$ at a distance of 5 units from it.

→ Soln,

let a line parallel to $5x + 12y = 0$ be

$$5x + 12y + k = 0$$

distance between parallel lines = 5 units

Here,

$$a = 5, b = 12, C_1 = 0, C_2 = k$$

Now,

distance between parallel lines

$$= \left| \frac{C_1 - C_2}{\sqrt{a^2 + b^2}} \right|$$

$$\text{or, } 5 = \left| \frac{0 - k}{\sqrt{5^2 + 12^2}} \right|$$

$$\text{or, } 5 = \left| \frac{-k}{\sqrt{25 + 144}} \right|$$

$$\text{or, } 5 = \pm \frac{k}{13}$$

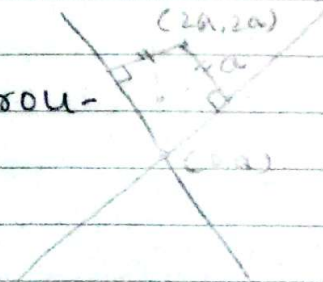
$$\text{or, } k = \pm 65$$

$\therefore$ eqn. of line is $5x + 12y \pm 65 = 0$

8. Find the equations of two straight lines drawn through the point $(0, a)$ on which the perpendiculars drawn from the point $(2a, 2a)$ are each of length a .

→ Soln,

The equation of line through the point $(0, a)$ is



$$y - y_1 = m(x - x_1)$$

$$\text{or, } y - a = m(x - 0)$$

$$\text{or, } y - a = mx$$

$$\text{or, } mx - y + a = 0 \rightarrow (i)$$

length of perpendicular drawn from the point $(2a, 2a)$ to the line (i) is a . so,

$$\text{length of perpendicular} = \pm \frac{mx_1 - y_1 + a}{\sqrt{m^2 + (-1)^2}}$$

$$a = \pm \frac{m \cdot 2a - 2a + a}{\sqrt{m^2 + 1}}$$

$$\text{or, } a = \pm \frac{m \cdot 2a - 2a + a}{\sqrt{m^2 + 1}}$$

$$\text{or, } a\sqrt{m^2 + 1} = \pm 2am - a$$

$$\text{or, } \cancel{a}\sqrt{m^2 + 1} = \pm \cancel{a}(2m - 1)$$

Squaring on both side,

$$m^2 + 1 = (2m - 1)^2$$

$$\text{or, } m^2 + 1 = (2m)^2 - 2 \cdot 2m \cdot 1 + 1$$

$$\text{or, } m^2 + \cancel{1} = 4m^2 - 4m + \cancel{1}$$

$$\text{or, } 4m^2 - m^2 - 4m = 0$$

$$\text{or, } 3m^2 - 4m = 0$$

$$\text{or, } m(3m - 4) = 0$$

Either,

$$m=0 \quad 3m-4=0$$

$$m = \frac{4}{3}$$

$$\therefore m = 0, \frac{4}{3}$$

Now,

Putting the value of m in eqn (i)

When $m=0$

$$mx - y + a = 0$$

$$\text{or, } 0 \cdot x - y + a = 0$$

$$\text{or, } y - a = 0 \rightarrow (ii)$$

When $m = \frac{4}{3}$

$$mx - y + a = 0$$

$$\text{or, } \frac{4}{3}x - y + a = 0$$

$$\text{or, } 4x - 3y + 3a = 0 \rightarrow (iii)$$

~~Eqn~~ Required equations are,

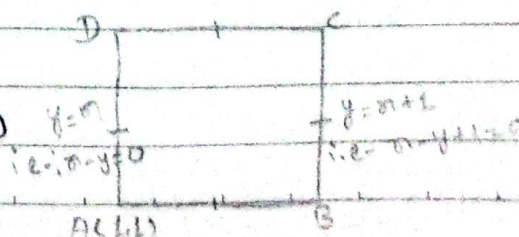
$$y - a = 0 \text{ and}$$

$$4x - 3y + 3a = 0 \downarrow$$

9. A vertex of a square is at $(1,1)$ and its two parallel sides are $y=x$ and $y=x+1$. Find the equations of the remaining side.

→ Soln,

Consider a square ABCD having co-ordinates of



At (1,1), equation of AD is $x=y$

$$\text{i.e. } x-y=0$$

and equation of BC is $y=x+1$

$$\text{i.e. } x-y+1=0$$

Now,

AB is perpendicular to AD so eqn of

AB is, $x+y+k=0$

Putting the value of x and y as (1,1)

$$x+y+k=0$$

$$\text{or, } 1+1+k=0$$

$$\text{or, } 2+k=0$$

$$\text{or, } k=-2$$

$\therefore$ Equation of AB is $x+y-2=0$

Similarly,

CD is perpendicular to AD and BC.

So, equation of CD is $x+y+c=0$

putting the value of x as $y-1$ which is the intersecting point,

$$y-1$$

10. If P and q are the lengths of perpendicular drawn from the point $(\cos \theta, \sin \theta)$ and $(-\sec \theta, \operatorname{cosec} \theta)$ on the line $x \sec \theta + y \operatorname{cosec} \theta = 0$ respectively, prove that $4p^2 + q^2 = 4$.

→ Soln,

P = length of perpendicular drawn from $(\cos \theta, \sin \theta)$ to the line

$$\begin{aligned} & x \sec \theta + y \operatorname{cosec} \theta = 0 \\ & = \pm \frac{\cos \theta \times \sec \theta + \sin \theta \times \operatorname{cosec} \theta}{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}} \\ & = \pm \frac{1 + 1}{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}} \end{aligned}$$

Squaring on both sides, we get

$$P^2 = \frac{4}{\sec^2 \theta + \operatorname{cosec}^2 \theta}$$

$$\text{or, } P^2 = \frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}$$

$$\text{or, } P^2 = \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cdot \cos^2 \theta}$$

$$\text{or, } P^2 = \frac{4 \sin^2 \theta \cdot \cos^2 \theta}{1}$$

$$\text{or, } P^2 = 4 \sin^2 \theta \cdot \cos^2 \theta$$

And,

q = length of perpendicular from $(-\sec \theta, \operatorname{cosec} \theta)$ to the line $x \sec \theta + y \operatorname{cosec} \theta = 0$

$$= \pm \frac{(-\sec \theta) \cdot \sec \theta + \operatorname{cosec} \theta \cdot \operatorname{cosec} \theta}{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}}$$

$$= \frac{\sec \theta \pm \operatorname{cosec}^2 \theta - \sec^2 \theta}{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}}$$

Squaring on both side, we get

$$q^2 = \frac{(\operatorname{cosec}^2 \theta - \sec^2 \theta)^2}{\sec^2 \theta + \operatorname{cosec}^2 \theta}$$

$$\begin{aligned} \text{or, } q^2 &= \left(\frac{\cos^2 \theta - \sin^2 \theta}{\sin^2 \theta \cdot \cos^2 \theta} \right)^2 \\ &= \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cdot \cos^2 \theta} \\ &= \frac{\cos^2 2\theta}{\sin^2 \theta \cdot \cos^2 \theta} \end{aligned}$$

Now,

$$L.H.S = 4p^2 + p^2 q^2$$

$$= 4 \times 4 \sin^2 \theta \cdot \cos^2 \theta + 4 \sin^2 \theta \cdot \cos^2 \theta \times \frac{\cos^2 2\theta}{\sin^2 \theta \cdot \cos^2 \theta}$$

$$= 4 (4 \sin^2 \theta \cdot \cos^2 \theta + \cos^2 2\theta)$$

$$= 4 ((2 \sin \theta \cdot \cos \theta)^2 + \cos^2 2\theta)$$

$$= 4 [\sin^2 2\theta + \cos^2 2\theta]$$

$$= 4 \times 1$$

$$= 4$$

Proved

Q. If P and q are the length of perpendicular drawn from origin to the lines $x \sec \theta - y \operatorname{cosec} \theta = a$ and $x \sin \theta + y \cos \theta = a \cos 2\theta$ respectively then prove that $4p^2 + q^2 = a^2$.

→ Soln,

P = length of perpendicular drawn from (0,0) to the line,

$$x \sec \theta - y \operatorname{cosec} \theta = a$$

$$P = \pm \frac{0 \sec \theta - 0 \operatorname{cosec} \theta - a}{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}}$$

$$= \pm \frac{-a}{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}}$$

Squaring on both side,

$$\text{or, } P^2 = \frac{a^2}{\sec^2 \theta + \operatorname{cosec}^2 \theta}$$

$$\text{or, } P^2 = \frac{a^2}{1}$$

$$\text{or, } P^2 = a^2 \sin^2 \theta \cdot \cos^2 \theta$$

And,

q = Length of perpendicular from origin $(0,0)$ to the line $x \sin \theta + y \cos \theta = a \cos 2\theta$

$$= \pm \frac{x \sin \theta + y \cos \theta - a \cos 2\theta}{\sqrt{\sin^2 \theta + \cos^2 \theta}}$$

$$= \pm \frac{0 \cdot \sin \theta + 0 \cdot \cos \theta - a \cos 2\theta}{\sqrt{\sin^2 \theta + \cos^2 \theta}}$$

$$= \pm \frac{a \cos 2\theta}{\sqrt{\sin^2 \theta + \cos^2 \theta}}$$

Squaring on both side,

$$q^2 = \frac{a^2 \cos^2 2\theta}{\sin^2 \theta + \cos^2 \theta}$$

$$\text{or, } q^2 = \frac{a^2 \cos^2 2\theta}{1}$$

Now,

$$\begin{aligned}
 \text{L.H.S} &= 4p^2 + q^2 \\
 &= 4(a^2 \sin^2 \theta \cdot \cos^2 \theta) + a^2 \cos^2 2\theta \\
 &= 4 \sin^2 \theta \cdot \cos^2 \theta \times a^2 + a^2 \cos^2 2\theta \\
 &= (2 \sin \theta \cdot \cos \theta)^2 \times a^2 + a^2 \cos^2 2\theta \\
 &= \sin^2 2\theta \times a^2 + a^2 \cos^2 2\theta \\
 &= a^2 (\sin^2 2\theta + \cos^2 2\theta) \\
 &= a^2 (1) [\because \sin^2 2\theta + \cos^2 2\theta = 1] \\
 &= a^2
 \end{aligned}$$

$$\therefore 4p^2 + q^2 = a^2$$

Proved

11 Show that the product of perpendiculars drawn from $(0, \pm \sqrt{b^2 - a^2})$, upon the line $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$ is a^2 .

→ Given, equation of line is:

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

$$\text{or, } b \cos \theta x + a \sin \theta y - ab = 0 \rightarrow (i)$$

let,

P = length of perpendicular from $(0, \sqrt{b^2 - a^2})$ to the line (i)

$$= \pm \frac{0 + a \sin \theta \sqrt{b^2 - a^2} - ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

$$= \pm \frac{a (\sin \theta \sqrt{b^2 - a^2} - b)}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

Ans, q = length of perpendicular from $(0, -\sqrt{b^2 - a^2})$ to the line (i)

$$= \pm \frac{0 - a \sin \theta \sqrt{b^2 - a^2} - ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

$$= \pm \frac{-a (\sin \theta \sqrt{b^2 - a^2} + b)}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

Now,

$$pq = \pm \frac{a (\sin \theta \sqrt{b^2 - a^2} - b)}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \times \frac{-a (\sin \theta \sqrt{b^2 - a^2} + b)}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

$$= \pm \frac{-a^2 [\sin^2 \theta (b^2 - a^2) - b^2]}{b^2 \cos^2 \theta + a^2 \sin^2 \theta}$$

$$= \pm \frac{-a^2 \sin^2 \theta (b^2 - a^2) + a^2 b^2}{b^2 (1 - \sin^2 \theta) + a^2 \sin^2 \theta}$$

$$= \pm \frac{-a^2 b^2 \sin^2 \theta + a^4 \sin^2 \theta + a^2 b^2}{b^2 - b^2 \sin^2 \theta + a^2 \sin^2 \theta}$$

$$= \pm \frac{a^2 (-b^2 \sin^2 \theta + a^2 \sin^2 \theta + b^2)}{(-b^2 \sin^2 \theta + a^2 \sin^2 \theta + b^2)}$$

$$= a^2$$

Two sides of a line

Consider a line

$$L: ax + by + c = 0$$

and two points $P(x_1, y_1)$ and $Q(x_2, y_2)$

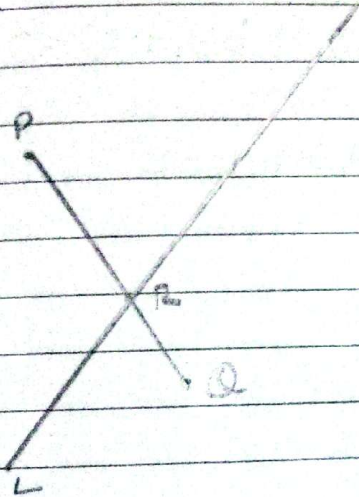


fig (i)



fig (ii)

Let the line segment joining the points P and Q intersect the line L at point R, produce PQ if necessary. Suppose R divides PQ in the ratio of $m_1:m_2$, then

$$\text{Coordinates of R} = \left(\frac{m_1 \cdot x_2 + m_2 \cdot x_1}{m_1 + m_2}, \frac{m_1 \cdot y_2 + m_2 \cdot y_1}{m_1 + m_2} \right)$$

This point also lies on the line L so,
 $a \left(\frac{m_1 \cdot x_2 + m_2 \cdot x_1}{m_1 + m_2} \right) + b \left(\frac{m_1 \cdot y_2 + m_2 \cdot y_1}{m_1 + m_2} \right) + c = 0$

$$\text{or, } \frac{am_1x_2 + am_2x_1}{m_1 + m_2} + \frac{bm_1y_2 + bm_2y_1}{m_1 + m_2} + c = 0$$

$$\text{or, } \frac{am_1x_2 + am_2x_1 + bm_1y_2 + bm_2y_1 + cm_1 + cm_2}{m_1 + m_2} = 0$$

$$\text{or, } m_1(ax_2 + by_2 + c) + m_2(ax_1 + by_1 + c) = 0$$

$$\text{or, } m_1(ax_2 + by_2 + c) = -m_2(ax_1 + by_1 + c)$$

$$\text{or, } \frac{m_1}{m_2} = - \frac{ax_1 + by_1 + c}{ax_2 + by_2 + c}$$

Conclusions:

① If $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ are of same sign, then $m_1:m_2$ is negative and hence the division is external. so, the points lies on the same side.

② If $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ are of opposite sign, then $m_1:m_2$ is $\oplus$ ve and hence the division is internal. so, the point lies on opposite side.

Q. Test whether the points (1,2) and (-2,3) lies on the same or opposite sides of the line $5x - 2y = 2$.

→ Soln,

Given line is

$$5x - 2y = 2$$

or, $5x - 2y - 2 = 0$ — (i)

For the point $(x_1, y_1) = (1, 2)$

$$\begin{aligned} 5x_1 - 2y_1 - 2 &= 5 \times 1 - 2 \times 2 - 2 \\ &= 5 - 6 \\ &= -1 \end{aligned}$$

For the point $(x_2, y_2) = (-2, 3)$

$$\begin{aligned} 5x_2 - 2y_2 - 2 &= 5 \times (-2) - 2 \times 3 - 2 \\ &= -10 - 6 - 2 \\ &= -18 \end{aligned}$$

As the signs are same, given points lies on the same side of the given line.

(12) Are $(1, 2)$ and $(4, -5)$ lies on the same side of the line $2x - 3y = 1$? Justify with calculation.

→ Soln,

Given line is

$$2x - 3y = 1$$

or, $2x - 3y - 1 = 0$

For the point $(x_1, y_1) = (1, 2)$

$$\begin{aligned} 2x_1 - 3y_1 - 1 &= 2 \times 1 - 3 \times 2 - 1 \\ &= 2 - 6 - 1 \\ &= 2 - 7 \\ &= -5 \end{aligned}$$

For the point $(x_2, y_2) = (4, -5)$

$$2x_2 - 3y_2 - 1 = 2 \times 4 - 3 \times (-5) - 1$$

$$\begin{aligned}
 &= 8 + 15 - 1 \\
 &= 8 + 14 \\
 &= 22
 \end{aligned}$$

As the signs are opposite, given point does not lie on same side of the line.

(13) Is the line $3x + 4y - 8 = 0$ lies between the origin and $(2, 4)$? Justify.

→ Soln,

Given line is,

$$3x + 4y - 8 = 0$$

$$\text{or, } 3x + 4y - 8 = 0$$

For

For the point $(x_1, y_1) = (0, 0)$

$$\begin{aligned}
 3x_1 + 4y_1 - 8 &= 3 \times 0 + 4 \times 0 - 8 \\
 &= -8
 \end{aligned}$$

For the point $(x_2, y_2) = (2, 4)$

$$\begin{aligned}
 3x_2 + 4y_2 - 8 &= 3 \times 2 + 4 \times 4 - 8 \\
 &= 6 + 16 - 8 \\
 &= 6 + 8 \\
 &= 14
 \end{aligned}$$

As the signs are opposite, given points lie on opposite side hence the given line lies between the origin and $(2, 4)$.

- (14) Find the locus which is equidistant from the lines $x-2y=0$ and $2y-11x=6$.

→ Soln,

Given two lines are,

$$x-2y=0 \rightarrow (i)$$

And

$$2y-11x=6$$

$$\text{i.e. } 11x-2y+6=0 \rightarrow (ii)$$

Let $P(x,y)$ be a moving point which is equidistant from the given lines.

So,

$$\frac{x-2y}{\sqrt{1^2+(-2)^2}} = \pm \frac{11x-2y+6}{\sqrt{11^2+(-2)^2}}$$

$$\text{or, } \frac{x-2y}{\sqrt{5}} = \pm \frac{11x-2y+6}{\sqrt{125}}$$

$$\text{or, } \frac{x-2y}{\sqrt{5}} = \pm \frac{11x-2y+6}{5\sqrt{5}}$$

$$\text{or, } 5x-10y = \pm (11x-2y+6)$$

Taking positive sign,

$$5x-10y = 11x-2y+6$$

$$\text{or, } 11x-5x-2y+10y+6=0$$

$$\text{or, } 6x+8y+6=0$$

$$\text{or, } 3x+4y+3=0$$

Taking -ve sign

$$5x-10y = -11x+2y-6$$

$$\text{or, } 11x+5x-10y-2y+6=0$$

$$\text{or, } 16x - 12y + 6 = 0$$

$$\text{or, } 8x - 6y + 3 = 0$$

$\therefore$ The required loci are $3x + 4y + 3 = 0$
and $8x - 6y + 3 = 0$.

Equation of Angle Bisectors:

Consider two straight lines

$$L_1: A_1x + B_1y + C_1 = 0$$

$$\text{and } L_2: A_2x + B_2y + C_2 = 0$$

Suppose B_1 and B_2 are the bisectors of the angle between the given lines. To

find the equation of these bisectors let us take a point $P(x, y)$ on any of the bisectors, say B_1 .

We know that, the bisector is at equal distance from its arms. so

$$PQ = PR \text{ where } PQ \perp L_1 \text{ and } PR \perp L_2$$

$$\text{or, } \frac{A_1x + B_1y + C_1}{\sqrt{A_1^2 + B_1^2}} = \pm \frac{A_2x + B_2y + C_2}{\sqrt{A_2^2 + B_2^2}}$$

This equation is true for any point $P(x, y)$ on the bisectors. So, the equations of bisectors are,

$$\frac{A_1x + B_1y + C_1}{\sqrt{A_1^2 + B_1^2}} = \pm \frac{A_2x + B_2y + C_2}{\sqrt{A_2^2 + B_2^2}}$$

And, $\frac{A_1x + B_1y + C_1}{\sqrt{A_1^2 + B_1^2}} = - \frac{A_2x + B_2y + C_2}{\sqrt{A_2^2 + B_2^2}}$

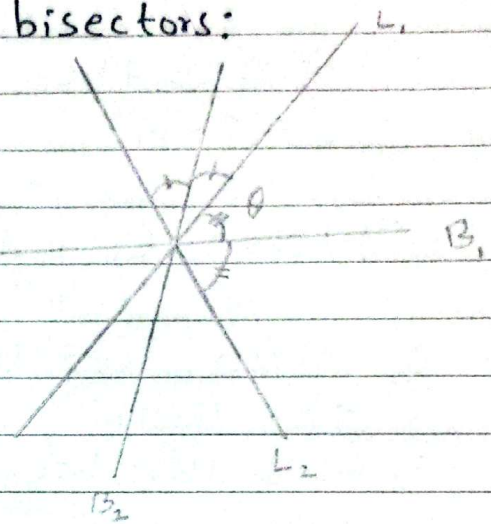
Identification of Angle bisectors:

① Acute and obtuse angle bisectors:

Let us take any one of the lines, say L_1 and any one of the bisectors, say B_1 . Suppose θ be the angle between L_1 and B_1 .

Then,

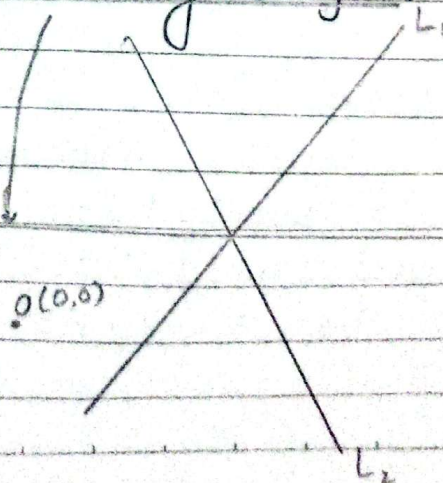
- if $|\tan \theta| > 1$, then the selected bisector is obtuse angle bisector and the other bisector B_2 is acute angle bisector.
- if $|\tan \theta| < 1$, then the selected bisector B_1 is acute angle bisector and the other bisector B_2 is the obtuse angle bisector.



② Bisector of the angle containing origin

Let the constant terms C_1 and C_2 of both equations are of same sign.

Then bisector of the angle containing origin is given by



$$\frac{A_1x + B_1y + C_1}{\sqrt{A_1^2 + B_1^2}} = \frac{A_2x + B_2y + C_2}{\sqrt{A_2^2 + B_2^2}} \text{ (Taking +ve)}$$

And, the bisector of angle not containing the origin is given by

$$\frac{A_1x + B_1y + C_1}{\sqrt{A_1^2 + B_1^2}} = -\frac{A_2x + B_2y + C_2}{\sqrt{A_2^2 + B_2^2}} \text{ (Taking -ve)}$$

Where C_1 and C_2 are of same sign.

Q Given Two lines $3x + 2y + 5 = 0$ and $2x + 3y = 8$. Then

i) Find the bisectors of angles between the lines.

ii) Show that the bisectors are at right angles.

iii) Identify the acute and obtuse angle bisectors.

iv) Identify bisector of the angle containing origin.

→ Soln,

Given lines are,

$$3x + 2y + 5 = 0 \rightarrow (i)$$

$$\text{and } 2x + 3y - 8 = 0$$

$$\text{i.e. } -2x - 3y + 8 = 0 \rightarrow (ii)$$

① The bisectors of angle between the lines are given by

$$\frac{3x+2y+5}{\sqrt{3^2+2^2}} = \pm \frac{-2x-3y+8}{\sqrt{(-2)^2+(-3)^2}}$$

$$\text{or, } \frac{3x+2y+5}{\sqrt{9+4}} = \pm \frac{-2x-3y+8}{\sqrt{4+9}}$$

$$\text{or, } \frac{3x+2y+5}{\sqrt{13}} = \pm \frac{-2x-3y+8}{\sqrt{13}}$$

$$\text{or, } 3x+2y+5 = \pm(-2x-3y+8)$$

Taking +ve sign,

$$3x+2y+5 = -2x-3y+8$$

$$\text{or, } 3x+2x+2y+3y+5-8=0$$

$$\text{or, } 5x+5y-3=0 \rightarrow \text{(iii)}$$

Taking -ve sign

$$3x+2y+5 = 2x+3y-8$$

$$\text{or, } 3x-2x+2y-3y+5+8=0$$

$$\text{or, } x-y+13=0 \rightarrow \text{(iv)}$$

$\therefore$ The bisectors are $5x+5y-3=0$ and $x-y+13=0$.

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⑪ Soln,

$$\begin{aligned}\text{Slope of eqn (iii)} [m_1] &= -\frac{x \text{ coe.}}{y \text{ coe.}} \\ &= \frac{-5}{5} \\ &= -1\end{aligned}$$

$$\begin{aligned}\text{Slope of eqn (iv)} [m_2] &= -\frac{x \text{ coe.}}{y \text{ coe.}} \\ &= \frac{-1}{-1} \\ &= 1\end{aligned}$$

Now,

$$\begin{aligned}m_1 \cdot m_2 &= -1 \times 1 \\ &= -1\end{aligned}$$

Since $m_1 \cdot m_2 = -1$, So the bisectors are at right angled. $\perp$.

⑫

Let θ be the angle between line (i) and bisector (iii)

For the line (i) i.e. $3x+2y+5=0$

$$\text{slope } (m_1) = -\frac{3}{2}$$

For the bisector (iii) i.e. $5x+5y-3=0$

$$\text{Slope } (m_2) = -\frac{5}{5} = -1$$

We have

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 \cdot m_2}$$

$$= \frac{-\frac{3}{2} - (-1)}{1 + \left(-\frac{3}{2}\right) \cdot (-1)}$$

$$= \frac{-\frac{3}{2} + 1}{1 + \frac{3}{2}}$$

$$= \frac{-\frac{1}{2}}{\frac{5}{2}}$$

$$= -\frac{1}{5}$$

i.e. $|\tan \theta| = \frac{1}{5} < 1$,

So the selected bisector is $5x + 5y - 3 = 0$ is the acute angle bisector and hence the remaining bisector i.e. $x - y + 13 = 0$ is the obtuse angle bisector.

(iv) The constant terms C_1 and C_2 of both the lines are of same sign. So, bisector of the angle containing origin is given by taking +ve sign.

$\therefore 5x + 5y - 3 = 0$ is a bisector of the angle containing origin.